



Finding Critical Numbers of Functions

Calculus Worksheet · Grade 11-12

Name: _____

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Learning Objectives

- Compute the first derivative of rational and polynomial functions
- Identify values where the derivative equals zero or is undefined
- Determine valid critical numbers by checking the domain of the original function

Find all critical numbers of each function by computing the first derivative and applying both conditions: $f'(c) = 0$ and $f'(c)$ undefined.

1. Find all critical numbers of the function.

$$f(x) = \frac{x^2}{x^2 - 4}$$

Answer: _____

2. Find all critical numbers of the polynomial.

$$f(x) = x^3 - 3x^2 - 9x + 5$$

Answer: _____

3. Find all critical numbers of the rational function.

$$f(x) = \frac{x}{x^2 + 1}$$

Answer: _____

4. Find all critical numbers of the function.

$$f(x) = x^4 - 8x^2 + 16$$

Answer: _____

5. Find all critical numbers of the radical function.

$$f(x) = \sqrt{x^2 - 9}$$

Answer: _____

6. Find all critical numbers of the function.

$$f(x) = \frac{x^2 - 1}{x^2 + 1}$$

Answer: _____



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7. Find all critical numbers of the function.

$$f(x) = x^{2/3}(x - 5)$$

Answer: _____

8. Find all critical numbers of the trigonometric function on $[0, 2\pi]$.

$$f(x) = \sin x + \cos x$$

Answer: _____

9. Find all critical numbers of the function.

$$f(x) = \frac{1}{x^2 - 1}$$

Answer: _____

10. Find all critical numbers of the exponential-polynomial function.

$$f(x) = xe^{-x}$$

Answer: _____





Remind students that a candidate value is only a valid critical number if it lies in the domain of the original function $f(x)$.

Solutions

1. Find all critical numbers of the function.

$$f(x) = \frac{x^2}{x^2 - 4}$$

- Use the quotient rule to differentiate $f(x)$.
- Simplify the derivative to negative $8x$ over the quantity x squared minus 4 , squared.
- Set the numerator equal to zero to get x equals 0 .
- Set the denominator equal to zero to get x equals plus or minus 2 .
- Check the domain of $f(x)$; x equals plus or minus 2 are excluded, so only x equals 0 is a valid critical number.

Answer: $x = 0$

2. Find all critical numbers of the polynomial.

$$f(x) = x^3 - 3x^2 - 9x + 5$$

- Differentiate to obtain f prime of x equals $3x$ squared minus $6x$ minus 9 .
- Factor as 3 times the quantity x squared minus $2x$ minus 3 .
- Factor further into 3 times x minus 3 times x plus 1 .
- Set each factor equal to zero to find x equals 3 and x equals negative 1 .

Answer: $x = -1, x = 3$

3. Find all critical numbers of the rational function.

$$f(x) = \frac{x}{x^2 + 1}$$

- Apply the quotient rule to get f prime of x equals the quantity 1 minus x squared over the quantity x squared plus 1 , squared.
- Set the numerator 1 minus x squared equal to zero.
- Solve to get x equals plus or minus 1 .
- The denominator is never zero, so both values are valid critical numbers.

Answer: $x = -1, x = 1$

4. Find all critical numbers of the function.

$$f(x) = x^4 - 8x^2 + 16$$

- Differentiate to get f prime of x equals $4x$ cubed minus $16x$.
- Factor as $4x$ times the quantity x squared minus 4 .
- Factor further into $4x$ times x minus 2 times x plus 2 .
- Setting each factor equal to zero gives x equals 0 , x equals 2 , and x equals negative 2 .

Answer: $x = -2, x = 0, x = 2$



5. Find all critical numbers of the radical function.

$$f(x) = \sqrt{x^2 - 9}$$

- Differentiate using the chain rule to get f' of x equals x over the square root of the quantity x squared minus 9.
- Set the numerator equal to zero to get x equals 0, but 0 is not in the domain of f .
- Set the denominator equal to zero to get x equals plus or minus 3.
- Since plus or minus 3 are endpoints of the domain, they are the critical numbers.

Answer: $x = \pm 3$

6. Find all critical numbers of the function.

$$f(x) = \frac{x^2 - 1}{x^2 + 1}$$

- Apply the quotient rule to obtain f' of x equals $4x$ over the quantity x squared plus 1, squared.
- Set the numerator equal to zero to get x equals 0.
- The denominator x squared plus 1 is never zero, so there are no undefined points.
- The only critical number is x equals 0.

Answer: $x = 0$

7. Find all critical numbers of the function.

$$f(x) = x^{2/3}(x - 5)$$

- Rewrite $f(x)$ as x to the five-thirds minus 5 times x to the two-thirds.
- Differentiate to get f' of x equals five-thirds times x to the two-thirds minus ten-thirds times x to the negative one-third.
- Combine over a common denominator to get the quantity $5x$ minus 10 over 3 times the cube root of x .
- Set the numerator equal to zero to get x equals 2, and the denominator equal to zero to get x equals 0.
- Both values lie in the domain of f , so they are critical numbers.

Answer: $x = 0, x = 2$

8. Find all critical numbers of the trigonometric function on $[0, 2\pi]$.

$$f(x) = \sin x + \cos x$$

- Differentiate to get f' of x equals cosine x minus sine x .
- Set f' of x equal to zero to get cosine x equals sine x .
- Divide both sides by cosine x to get tangent x equals 1.
- On the interval from 0 to 2π , the solutions are x equals $\pi/4$ and x equals $5\pi/4$.

Answer: $x = \frac{\pi}{4}, x = \frac{5\pi}{4}$

9. Find all critical numbers of the function.

$$f(x) = \frac{1}{x^2 - 1}$$

- Differentiate to obtain f' of x equals negative $2x$ over the quantity x squared minus 1, squared.
- Set the numerator equal to zero to get x equals 0.
- Set the denominator equal to zero to get x equals plus or minus 1, but these are not in the domain of f .
- Therefore, x equals 0 is the only critical number.

Answer: $x = 0$



10. Find all critical numbers of the exponential-polynomial function.

$$f(x) = xe^{-x}$$

- Differentiate using the product rule to get f' prime of x equals e to the negative x minus x times e to the negative x .
- Factor out e to the negative x to obtain e to the negative x times the quantity 1 minus x .
- Since e to the negative x is never zero, set 1 minus x equal to zero.
- Solving gives x equals 1 as the only critical number.

Answer: $x = 1$

